VISUAL ARCHITECTURE & PROCESS BLUEPRINT

AI Personal Career Counselor

Visual Flowcharts, State Machine Models, Pipeline Architecture & Engineering Methodologies

DOCUMENT SCOPE	KEY STATE ENGINE	MATCHING STRATEGY	LAST UPDATED
Flowcharts & Workflows	ProtectedRoute Route Guards	Hybrid Multi-Factor Scoring	October 2026

1. Executive Summary: The Two Pillars of Guidance

The architecture of the **AI Personal Career Counselor** is built on two distinct lifecycle roles designed to guide students from career ambiguity to daily job readiness:

- **Pillar 1: Counselor Mode (Discovery & Validation)** — Uses psychometric Holland RIASEC profiling, subconscious identity tests ("Know Me"), resume extraction, and dense vector similarity across 1,016 O*NET occupations to determine optimal career alignment.
- **Pillar 2: Coach Mode (Execution & Habit Formation)** — Once a career is deliberately locked in `career_goals`, the application generates a phased roadmap, computes NLP skill gaps, dynamically schedules daily tasks, and tracks progress streaks.

> **Core Routing Rule: Choose First, Then Plan**
> The platform prevents premature access to the dashboard. Every student must complete Onboarding and Assessment, and explicitly commit to a single active career goal before Phase 2 coach modules unlock.

1.1 Journey Milestone Progression Summary

STAGE	PRIMARY COMPONENT	DATABASE STATE	NEXT PERMITTED ACTION
0. Auth	`Login.tsx` / `Signup.tsx`	Supabase Auth Session	Authenticate → Advance to Stage 1
1. Onboard	`Onboarding.tsx`	`profiles` record written	Submit → Navigate to `/assessment`
2. Assess	`Assessment.tsx`	`assessments` record written	Score → Navigate to `/assessment?step=results`
3. Confirm	`Assessment.tsx` modal	`career_goals` (status='active')	Confirm → Unlock `/dashboard` & Phase 2
4. Execute	`Dashboard.tsx` / `Today.tsx`	`roadmaps` & `task_progress`	Daily study tasks, skill gap & level completion

2. Flowchart 1: Student Journey & Route Guard State Machine

This flowchart illustrates the state progression enforced by `ProtectedRoute.tsx`, handling authentication, profile hydration, milestone gates, and automatic URL redirection:

1. Entry & Session Verification — Supabase Auth

Student visits platform → Check session. If unauthenticated, redirect to `/login` with return path saved in route state.

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2. State Hydration Guard — Anti-Loop Check

Wait for `authLoading == false` and `profileLoading == false`. Defer all routing decisions until database profile and latest assessment rows are resolved to prevent infinite redirect loops.

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3. Gate 1: Onboarding Evaluation — hasCompletedOnboarding()

Checks: Educational tier, course, graduation year, interests, weekly hours.
If Incomplete: Restrict to `/onboarding`. Attempts to visit `/assessment` or `/dashboard` redirect here.
If Complete: Advance to Assessment evaluation.

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4. Gate 2: Assessment Evaluation — latestAssessment != null

Checks: 24 Holland RIASEC questions scored + 'Know Me' subconscious preferences recorded.
If Incomplete: Restrict to `/assessment`. Premature access to `/dashboard` redirects to `/assessment`.
If Complete: Advance to Goal Commitment evaluation.

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5. Gate 3: Career Goal Confirmation — activeCareerGoal != null

Checks: Student has committed to one career in `career goals` (status = 'active') and synced with `profiles.target_role`.
If Not Confirmed: Redirect to `/assessment?step=results` to force deliberate goal selection.
If Confirmed: Full authorization granted.

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6. Career Mode Activated — Dashboard Unlocked

Unrestricted access to `/dashboard`, `/today`, `/roadmap`, `/skills`, and `/profile`.
Site personalizes: "*Good morning, future Software Engineer* 🍸", generates customized weekly timetable, and loads NLP skill gap.

3. Flowchart 2: Hybrid Multi-Factor Career Matching & AI Fallback Pipeline

Recommendation generation is a deterministic, multi-factor ranking pipeline that scores 1,016 O*NET occupations against student inputs, paired with an intelligent LLM reranker:

1. Multi-Dimensional Profile Inputs — Client Request

RIASEC scores (R, I, A, S, E, C vectors) • Extracted resume skills • Academic tier & course • "Know Me" subconscious values & energy sources • Anti-goal dealbreakers.

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50% Dense Vector Match
`all-MiniLM-L6-v2` embeddings. Cosine similarity between student background text and 1,016 O*NET occupation vectors.

30% RIASEC Fit
Normalized Euclidean distance between student's 6 Holland dimensions and target occupation's empirical profile.

20% Skill Overlap
Direct match ratio between parsed resume competencies and target occupation required skills dictionary.

Dealbreaker Penalties
Deductions (-0.35 per conflict) for non-negotiable anti-goals (e.g. "no coding", "no night shifts", "relocation budget").

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2. Aggregate Ranking & Top-15 Candidate Selection — Deterministic Vector Sort

Calculate weighted score: `score = 0.50*vector + 0.30*riasec + 0.20*skills - penalties`.
Rank all 1,016 occupations and filter the top 15 highest-scoring candidates for synthesis.

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Track A: Groq AI Synthesis (Primary)
Trigger: Groq API is healthy.
Engine: `openai/gpt-oss-120b` (JSON Mode).
Deliverable: Re-ranks top 5 careers, crafts personalized fit reasons, and identifies trade-offs & watch-outs.

Track B: Rule-Based Fallback (Offline)
Trigger: Rate limit, timeout, or AI service outage.
Engine: Deterministic category rules.
Deliverable: Selects top 5 purely by aggregate mathematical score with pre-cached archetype match reasons.

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3. Output Persistence & Client Delivery — assessments.matches

Atomic persistence to Supabase `assessments` table with `source = 'ai' | 'rule'`.
Returns top 5 recommendations with compatibility percentage, salary tiers, and personalized reasoning to the frontend.

4. Flowchart 3: Engineering Implementation Methodology (5 Phases)

The disciplined 5-phase engineering workflow used across Tasks 1–4 to eliminate regressions and guarantee production stability:

Phase 1: Codebase Audit & Contract Discovery — Zero Assumptions

Inspected Supabase schemas, Pydantic models, and React context hooks. Verified database column definitions and partial indexes before touching code.

Phase 2: State Modeling & Pure Predicates — Deterministic Logic

Extracted journey stage validation into pure functions (`hasCompletedOnboarding`, `hasCompletedAssessment`, `hasConfirmedGoal`) in `profile.ts`.

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Phase 3: Route Guarding & Atomic Persistence — Defensive Implementation

Rewrote `ProtectedRoute.tsx` with async hydration checks. Reused `confirmCareerGoal()` to write atomically to `career_goals` and `profiles.target_role`.

Phase 4: Automated Verification Testing — 8 Regression Suites

Simulated navigation, page refreshes, direct URL tampering, and Groq rate-limiting outages. Verified 100% test pass rate across all journeys.

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Phase 5: Automated Documentation & Print Engine — Production PRD

Built a headless browser compilation script using CSS Paged Media to produce publication-grade PDF specification documents.

5. Technical Architecture & Component Interaction

COMPONENT	SOURCE IMPLEMENTATION	ROLE & ARCHITECTURAL RESPONSIBILITY
Route Guard	`ProtectedRoute.tsx`	Enforces linear progression. Intercepts unauthorized navigation, handles loading states, and redirects returning students to incomplete steps.
Journey State	`profile.ts` & `ProfileContext.tsx`	Single source of truth for onboarding data, latest assessment, active career goal, and milestone verification predicates.
Matching Engine	`backend/matcher.py` & `careers.py`	Multi-factor hybrid scoring across 1,016 O*NET occupations using MiniLM vector embeddings, RIASEC distance, and skill overlap.
AI Reranker	`backend/main.py` (Groq LPU)	Contextual LLM reasoning producing tailored match rationale with deterministic mathematical fallback if offline.
Goal Persistence	`career_goals` (Supabase PostgreSQL)	Guaranteed by partial unique index `career_goals_one_active` ensuring exactly one active career goal per student.

6. Key Design Decisions & Trade-Off Analysis

CHALLENGE	ADOPTED SOLUTION	ARCHITECTURAL RATIONALE
Redirect Loops on Refresh	Multi-gate state hydration checks (`authLoading || profileLoading`)	Prevents routing decisions from firing while asynchronous Supabase user metadata and profile rows are still loading.
Goal Inconsistency	Atomic synchronization across `career_goals` and `profiles.target_role`	Eliminates data drift between the active goal entity and profile context, ensuring the dashboard always loads the correct target role.
AI Service Outages	Dual-track architecture (LLM Rerank + Deterministic Fallback)	Provides high availability. If the Groq API exceeds rate limits or is unreachable, the matching engine continues operating without errors.
Direct URL Bypass	Strict prerequisite milestone checking in route guards	Guarantees that unauthenticated or unassessed users cannot jump directly to `/dashboard` or `/roadmap` via manual URL entry.

7. Verification Matrix Summary

AUDIT TEST SCENARIO	TARGET COMPONENT	RESULT	OBSERVED PRODUCTION BEHAVIOR
Onboarding → Assessment Flow	`Onboarding.tsx` → `Assessment.tsx`	Verified	Completing onboarding navigates to assessment without premature dashboard redirect.
Goal Confirmation Persistence	`Assessment.tsx` → `career_goals`	Verified	Career title, slug, and field write atomically and satisfy the unique active goal constraint.
Journey Route Protection	`ProtectedRoute.tsx`	Verified	Unauthenticated or incomplete milestone attempts are redirected to their required step.
Browser Refresh State Retention	`ProfileContext.tsx`	Verified	Saved onboarding inputs, RIASEC answers, and confirmed goals rehydrate cleanly on refresh.
Offline Resilience & Fallback	`matcher.py` & `main.py`	Verified	Groq outage simulation confirms instant fallback to deterministic mathematical ranking.

Document Summary: This document serves as the visual and methodological blueprint for the AI Personal Career Counselor architecture, detailing the state machine, hybrid matcher, and 5-phase engineering methodology.